

Periodic Homogenization of Hamilton–Jacobi Equations for Infinite Systems of Indistinguishable Particles

From the general $O(\varepsilon^{1/3})$ rate to the optimal $O(\varepsilon)$ rate under convexity / Seho Park park646@wisc.edu

1 Setting and motivation

The continuum of indistinguishable particles on the $\mathbb{T}^d$. [Gomes and Nurbekyan]

- Particle labels: $i \in I = [0, 1]^d$. Lebesgue measure λ_0 on I
- Configuration: $V := L^2(I; \mathbb{R}^d)$. $x(i)$ = position of particle i
- The dynamics can be described by the Hamiltonian $H(x, p) : V \times V \rightarrow \mathbb{R}$

Infinite-dimensional oscillatory problem

$$(\text{CP})_\varepsilon \quad \begin{cases} u_t^\varepsilon + H\left(\frac{x}{\varepsilon}, Du^\varepsilon\right) = 0 & \text{in } V \times (0, \infty), \\ u^\varepsilon(x, 0) = u_0(x) & \text{on } V. \end{cases}$$

Symmetries of the problem

$$\Lambda := L^2(I; \mathbb{Z}^d), \quad \mathcal{G} := \{\text{measure preserving bijections of } I\}$$

$$\text{Periodicity: } H(x + z, p) = H(x, p) \quad \text{for } z \in \Lambda$$

$$\text{Indistinguishability: } H(x \circ g, p \circ g) = H(x, p) \quad \text{for } g \in \mathcal{G}$$

V is not locally compact. However, **the quotient space is compact**.

$$SS^d := V/\Lambda/\mathcal{G}, \quad d_{SS^d}(x, y) := \inf_{g \in \mathcal{G}, z \in \Lambda} \|x - y \circ g - z\|_{L^2}.$$

This replaces the local compactness used in finite-dimensional theory.

Mean-field example

$$H(x, p) = \frac{1}{2} \|p\|_{L^2}^2 + \int_I V_0(x(i)) d\lambda_0 + \frac{1}{2} \int_I W_0(x(i) - x(j)) d\lambda_0(i) d\lambda_0(j)$$

$V_0 \in W^{1,\infty}(\mathbb{T}^d)$: external potential / $W_0 \in W^{1,\infty}(\mathbb{T}^d)$: pairwise interaction

Assumptions

- H : Local Lipschitz continuity, and coercivity in p ($\lim_{\|p\| \rightarrow \infty} \inf_{x \in V} H(x, p) = +\infty$).
- u_0 : $u_0(x) = u_0(Mx)$ and $u_0 \in C_b^1(V)$, where

$$\mathfrak{m}(x) := \int_I x d\lambda_0 \in \mathbb{R}^d, \quad Mx := \mathfrak{m}(x)\chi_I.$$

Question. As $\varepsilon \downarrow 0$, does $u^\varepsilon \rightarrow u$? What is the equation of u ? How fast?

2 Main results

∞ -dim equation $\xrightarrow{\text{homogenization}}$ **finite-dimensional effective dynamics**.

Convergence rate: $O(\varepsilon^{1/3})$ for nonconvex H / $O(\varepsilon)$ for convex H

Under certain conditions,

$$u^\varepsilon(x, t) \longrightarrow \tilde{u}(\mathfrak{m}(x), t)$$

locally uniformly, where **the limit solves the finite-dimensional equation**

$$(\overline{\text{CP}}) \quad \begin{cases} \tilde{u}_t + \overline{H}(D\tilde{u}) = 0 & \text{in } \mathbb{R}^d \times (0, \infty), \\ \tilde{u}(q, 0) = \tilde{u}_0(q) & \text{on } \mathbb{R}^d. \end{cases}$$

Moreover, for every $T > 0$,

$$\sup_{V \times [0, T]} |u^\varepsilon(x, t) - \tilde{u}(\mathfrak{m}(x), t)| \leq C_T \varepsilon^{1/3}.$$

If $p \mapsto H(x, p)$ is **convex**, then

$$\|u^\varepsilon(x, t) - \tilde{u}(\mathfrak{m}(x), t)\|_{L^\infty(V \times [0, \infty))} \leq C\varepsilon.$$

The convergence rate $O(\varepsilon)$ is **optimal**.

3 Infinite-dimensional viscosity framework

Smooth test function ϕ touches u from **above** / **below** at (x_0, t_0) ,

$$\phi_t(x_0, t_0) + H(x_0, D\phi(x_0, t_0)) \leq 0 / \geq 0.$$

In infinite dimensions, a bounded continuous function **need not attain its max / min**.

Radon–Nikodym property

For every continuous function φ on a closed ball $B \subset V$ and every $\varepsilon > 0$, there exists $p \in V^*$ with $\|p\| \leq \varepsilon$ such that $\varphi + p$ attains its min / max on B .

These enable the infinite-dimensional test-function arguments. [Crandall and Lions]

Lipschitz regularity of u^ε

$$\text{Construct Barriers: } \psi^\pm(x, t) := u_0(x) \pm C_0 t$$

- $\psi^- \leq u^\varepsilon(x, t) \leq \psi^+ \Rightarrow \|u_t^\varepsilon\|_\infty \leq C_1$ (Lip in time)
- Coercivity implies $|H(x/\varepsilon, Du^\varepsilon)| \leq C_1 \Rightarrow \|Du^\varepsilon\|_\infty \leq C_2$ (Lip in x)

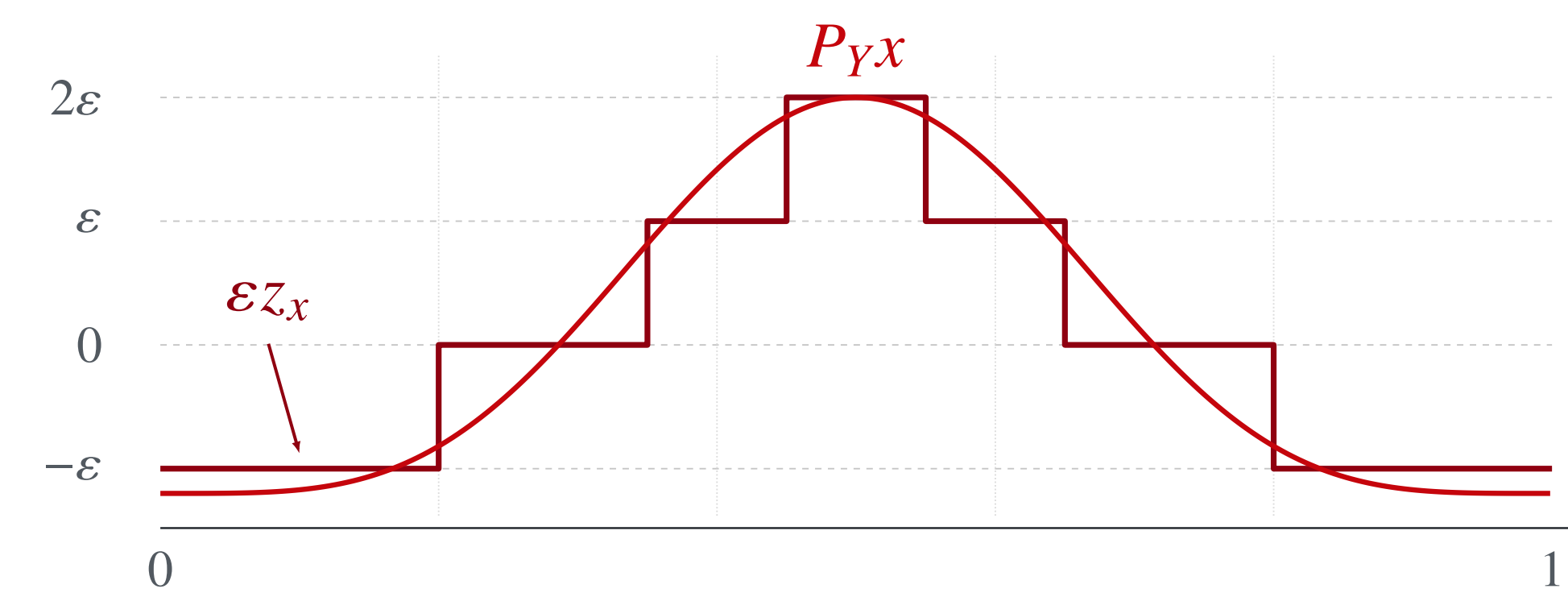
4 How symmetry produces the model reduction

4.1 Decompose the configuration space

$$Y := \{y \in V : \mathfrak{m}(y) = 0\}, \quad V = Y \oplus Y^\perp, \quad Y^\perp = \{q\chi_I : q \in \mathbb{R}^d\}.$$

Every $x \in V$ admits the microscopic decomposition

$$x = Mx + P_Y x = Mx + \varepsilon z_x + x_\varepsilon, \quad z_x \in \Lambda \cap Y, \quad \|x_\varepsilon\| \leq C\varepsilon.$$



Because $H(\frac{x+\varepsilon z}{\varepsilon}, p) = H(\frac{x}{\varepsilon}, p)$, u^ε is invariant under εz_x -translation. Thus,

$$|u^\varepsilon(x, t) - u^\varepsilon(Mx, t)| \leq C\varepsilon.$$

Only the mean component survives in the limit.

4.2 Ansatz and cell problem

$$u^\varepsilon(x, t) = \tilde{u}(\mathfrak{m}(x), t) + \varepsilon v(y), \quad y = x/\varepsilon,$$

separates the scales and produces

$$\begin{aligned} H(y, p\chi_I + Dv(y)) &= \overline{H}(p) & \text{in } V & \quad (\text{cell problem}) \\ \tilde{u}_t + \overline{H}(D\tilde{u}) &= 0 & \text{in } \mathbb{R}^d \times (0, \infty) & \quad (\text{effective equation}) \end{aligned}$$

Construct v through the discounted equation

$$\lambda v^\lambda + H(y, p\chi_I + Dv^\lambda) = 0.$$

- Transfer both symmetries to v^λ , so it **descends to compact SS^d** .
- Arzelà–Ascoli and $\lambda \downarrow 0$ give a corrector and the unique ergodic constant $\overline{H}(p)$.

4.3 Perturbed-test-function argument [Evans]

Doubling variables:

$$\Phi(x, y, t) = u^\varepsilon(x, t) - \phi(x, t) - \varepsilon v(y) - \frac{1}{\eta} \left\| y - \frac{x}{\varepsilon} \right\|^2.$$

- RNP gives the maximum $(\hat{x}, \hat{y}, \hat{t})$. Penalty term guarantees $y \approx x/\varepsilon$.
- Use viscosity inequalities from $(\text{CP})_\varepsilon$ and cell problem.

5 The optimal convergence under convexity

Optimal-control representation.

$$u^\varepsilon(x, t) = \inf_{\gamma \in AC([0, t/\varepsilon]; V)} \left\{ u_0(\varepsilon\gamma(0)) + \varepsilon \int_0^{t/\varepsilon} L(\gamma(\tau), \dot{\gamma}(\tau)) d\tau : \gamma(t/\varepsilon) = x/\varepsilon \right\}$$

Question. The large-time behavior of Lagrangian action.

5.1 Proof Idea: Mean-endpoint action metric

For $t > 0$ and $a, q \in \mathbb{R}^d$, define

$$m(t, a, q) := \inf_{\gamma \in AC([0, t]; V)} \left\{ \int_0^t L(\gamma(s), \dot{\gamma}(s)) ds : \mathfrak{m}(\gamma(0)) = a, \mathfrak{m}(\gamma(t)) = q \right\}.$$

Consider the mean-restricted value function

$$U^\varepsilon(q, t) := \inf_{\mathfrak{m}(x)=q} u^\varepsilon(x, t) = \inf_{a \in \mathbb{R}^d} \left\{ \tilde{u}_0(a) + \varepsilon m\left(\frac{t}{\varepsilon}, \frac{a}{\varepsilon}, \frac{q}{\varepsilon}\right) \right\}, \quad q \in \mathbb{R}^d.$$

Almost sub-additivity and super-additivity estimates:

$$m(2t, 0, 2q) \leq 2m(t, 0, q) + C_R, \quad 2m(t, 0, q) \leq m(2t, 0, 2q) + C_R.$$

Homogenized metric and its estimate:

$$\overline{m}(t, a, q) := \lim_{\rho \rightarrow \infty} \frac{1}{\rho} m(\rho t, \rho a, \rho q), \quad \left| \varepsilon m\left(\frac{t}{\varepsilon}, \frac{a}{\varepsilon}, \frac{q}{\varepsilon}\right) - \overline{m}(t, a, q) \right| \leq C_R \varepsilon.$$

Identify $\overline{m}$ with the effective Lagrangian:

$$\overline{m}(t, a, q) = t \overline{L}\left(\frac{q-a}{t}\right), \quad \overline{L} = \overline{H}^*.$$

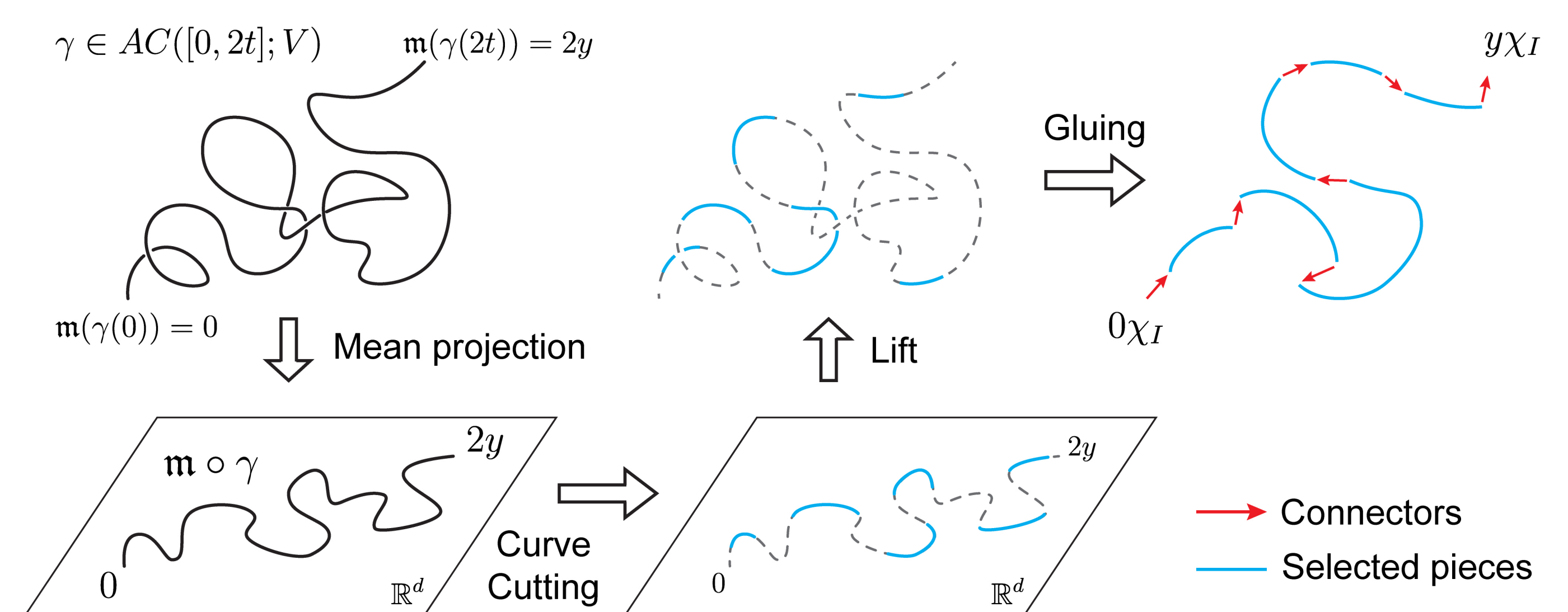
Hopf-Lax formula for the effective equation:

$$\tilde{u}(q, t) = \inf_{a \in \mathbb{R}^d} \left\{ \tilde{u}_0(a) + t \overline{L}\left(\frac{q-a}{t}\right) \right\} = \inf_{a \in \mathbb{R}^d} \left\{ \tilde{u}_0(a) + \overline{m}(t, a, q) \right\}, \quad q \in \mathbb{R}^d$$

Obtain optimal $O(\varepsilon)$ rate: $\|u^\varepsilon(x, t) - \tilde{u}(\mathfrak{m}(x), t)\|_{L^\infty(V \times [0, \infty))} \leq C\varepsilon$.

5.2 Infinite-dimensional curve surgery

From a minimizer γ of $m(2t, 0, 2y)$, construct a competitor for $m(t, 0, y)$.



- Apply Burago's lemma to the finite-dimensional mean–time path.
- Lift the pieces to V , then $\mathcal{G}$ -rearrange and Λ -translate them (L does not change).
- Compactness of SS^d** gives uniform bounds on the actions of the connectors.

Summary

- Symmetry buys compactness:** Symmetry of H gives the model reduction, while the compact quotient SS^d yields the corrector and qualitative homogenization.
- Convexity buys sharpness:** The optimal-control formula and a mean-endpoint action metric, combined with an ∞ -dim curve-surgery, sharpen the rate from $O(\varepsilon^{1/3})$ to the optimal $O(\varepsilon)$.